ARUNJYOTI SARMA

Doctoral Research Scholar | Indian Institute of Technology (BHU), Varanasi

FPGA Systems Architect • Flexible Electronics & Micro-Sensors • Embedded Systems & Edge AI

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1. EXECUTIVE SUMMARY & RESEARCH PROFILE

Hardware Systems Engineer and Doctoral Scholar at the Department of Electronics Engineering, **Indian Institute of Technology (BHU) Varanasi**, specializing in flexible electronics, strain-sensing structural health monitoring (SHM), and ultra-low-latency FPGA digital architectures. Demonstrated track record of translating advanced multi-physics modeling and analog sensor conditioning into verified silicon implementations: architected and physically verified sub-microsecond FPGA limit order book engines (322 MHz, 0 DSP, 82 mW power), fault-tolerant neuromorphic co-processors with autonomous silicon BIST (NeuroBIST-Edge), post-quantum lattice cryptography hardware units (TRISHUL), and flexible PVDF piezoelectric biomechanical energy harvesting (lead author, IEEE-ECBIOS 2026). Combines 8+ years of dedicated self-directed engineering R&D;, sustained national competitive mastery (four-time GATE qualifier in EC & EE: 2019, 2021, 2023, 2024), and advanced doctoral research at India's foremost premier institutions.

2. EDUCATION & ACADEMIC QUALIFICATIONS

Ph.D. in Electronics Engineering (Doctoral Scholar) — *IIT (BHU), Varanasi* *July 2024 – Present*

- **Doctoral Research:** Flexible electronics, strain-sensing arrays for structural health monitoring (SHM), piezoelectric nanogenerators (PENG), analog front-end readout circuits, and embedded digital telemetry.
- **Completed Coursework (54 Total Credits):** Signal Processing (EC-511), Information Theory & Coding (EC-512), Neural Networks (EC-523), Microprocessor Engineering & Applications (EC-521), Mobile Communication & Wireless Networks (EC-513), Switching Theory & Logic Design (EC-522).
- **Fellowship Award:** Conferred full-time Institute Research Fellowship by the Ministry of Education (MoE), Government of India.

B.Tech. in Electronics & Electrical Engineering — *KIIT University, Bhubaneswar* *2012 – 2016*

- **Academic Standing:** CGPA: **7.51 / 10.0** (First Class with Distinction). Active Member, KIIT Robotics Society.
- **National GATE Qualifications:** Four-time national qualifier in the Graduate Aptitude Test in Engineering across both Electronics & Communication (EC) and Electrical Engineering (EE) — **GATE 2024, GATE 2023, GATE 2021, and GATE 2019**.
- **Undergraduate Capstone:** Industrial Induction Motor Control via DTMF Telemetry; Autonomous Renewable Energy Generation Systems.

3. PROFESSIONAL & ADVANCED ENGINEERING RESEARCH EXPERIENCE

Lead Hardware Systems R&D; Engineer (Independent Applied Research) — *Applied R&D; Laboratory* *June 2016 – June 2024*

- Conducted an 8-year sustained research program in digital RTL acceleration, embedded architectures, and hardware-software co-design; repeatedly validated foundational mastery by qualifying national GATE benchmarks (2019, 2021, 2023, 2024) across EC and EE branches.
- **Open-Source Silicon IP:** Architected and verified TRISHUL (reconfigurable lattice cryptography hardware unit for Spartan-7/ASIC), NeuroBIST-Edge (fault-tolerant spiking processor with self-healing BIST), and USR-X (gear-shift neural computing register IP).
- **Embedded Firmware & Telemetry:** Developed custom firmware for ARM Cortex, ATmega, and PIC microcontrollers for industrial motor drives, acoustic sensing, and wireless telemetry.
- **Technical Mentorship:** Mentored engineering scholars in Microprocessors, Power Electronics, Circuit Theory, Signals & Systems, and Digital Logic Design (2017 – 2023).

Doctoral Research Scholar — Department of Electronics Engineering — *IIT (BHU), Varanasi* *July 2024 – Present*

- Leading experimental synthesis, multiphysics modeling, and circuit integration of piezoelectric nanogenerators (PENG) and flexible thin-film strain sensors for structural health monitoring (SHM) and bioelectronics.
- Synthesized 100 µm SOLEF 6008 PVDF flexible polymer films with aligned beta-phase dipoles under 190 °C, 2-metric-ton hot-press crystallization, generating 1.1 Vpp biomechanical voltage at 1.5 Hz.
- Engineered 4-stage active power-conditioning hardware (Schottky bridge, RC smoothing, MT3608 high-frequency boost converter, 220 µF tank accumulating 79.5 µJ) to autonomously drive micro-sensors.
- Modeled multi-physical strain dynamics in COMSOL Multiphysics for serpentine interconnects on elastomeric substrates to balance stretchability, gauge factor, and electrical continuity.

4. KEY SILICON IMPLEMENTATIONS & HARDWARE RESEARCH PROJECTS

NeuroQuant-LOB: Sub-Microsecond FPGA Line-Rate Order Book & Inference Engine [*AMD Spartan-7 FPGA • Verilog HDL • C++20 • Python • WebSockets*] — github.com/arunjyotisarmarsece24-jpg/neuroquant-lob

- Architected a deterministic hardware-accelerated limit order book (LOB) closed on physical Digilent Arty S7-25 (Spartan-7 XC7S25) on COM9 alongside an Arduino Nano HIL sidecar on COM10.
- Achieved timing closure at 322.06 MHz wire-speed (WNS = +0.894 ns) using 0 DSP slices (100% fixed-point LUT arithmetic, consuming only 240 / 14,600 LUTs = 1.64% silicon footprint) and 82 mW power (3,049x efficiency vs Xeon).
- Certified sub-microsecond latency (P50 = 1.230 µs, P99 = 2.451 µs) beating institutional <10.0 µs SLAs; walk-forward Deflated Sharpe Ratio DSR = 1.0000 across 500 permutations.

NeuroBIST-Edge: Fault-Tolerant Neuromorphic Spiking Processor with Autonomous Silicon BIST [*AMD Spartan-7 XC7S25 • Vivado ILA • Verilog HDL • ESP8266 Wi-Fi*] — github.com/arunjyotisarmarsece24-jpg/NeuroBIST-Edge

- Engineered an autonomous fault-tolerant neuromorphic co-processor with integrated silicon Built-In Self-Test (BIST) and Mod-3 Residue Invariant self-healing datapath.
- Implemented real-time Vivado ILA hardware digital twin and asynchronous Clock Domain Crossing (CDC) telemetry ingestion via Espressif ESP8266 Wi-Fi bridge.
- Demonstrated autonomous on-chip fault detection, error classification, and sub-cycle state recovery under live adversarial fault injection.

TRISHUL: Trusted Reconfigurable Integrated Security Hardware Unit for Lattice Cryptography [*Spartan-7 FPGA • ASIC-Oriented RTL • Post-Quantum Cryptography • Verilog*] — github.com/arunjyotisarmarsece24-jpg/TRISHUL

- Architected an open-source parameterized, reconfigurable cryptographic compute architecture developed from microarchitecture to Spartan-7 FPGA implementation and ASIC-ready physical design.
- Designed dedicated hardware accelerators for post-quantum lattice-based polynomial multiplication and high-throughput modular arithmetic with side-channel mitigation.

VEGA RISC-V Processing-in-Memory AI Cancer Diagnostic Accelerator (DVCon India 2025) [*AMD Kintex-7 FPGA • Indigenous VEGA RISC-V Core • Systolic Array • AXI4*] — Stage 2 Finalist & Lead Architect (Team AstraVortex, IIT BHU)

- Architected a Processing-in-Memory (PIM) accelerator integrated with India's indigenous VEGA RISC-V processor on a Genesys-2 Kintex-7 FPGA board.
- Embedded a Systolic Matrix Multiplication Engine (MME) in BRAM arrays to eliminate von Neumann memory bottlenecks; executed INT8 CNN/ViT inference with <20 ms latency and <6.8 W peak power.

Architecting Modern BIST & USR-X Gear-Shift Computing Register IP [*Verilog HDL • Yosys • GTKWave • Python • Built-In Self-Test*] — github.com/arunjyotisarmarsece24-jpg/Architecting-Modern-Built-In-Self-Test

- Authored architecture-first BIST research platform integrating LFSR PRPG, MISR compaction, and segmented gear-shift Universal Shift Register IP (USR-X).
- Enabled dual-mode compute agility: operates as a bit-serial multiply-accumulate (MAC) core for MLPs/CNNs, an XNOR+popcount Binary Neural Network (BNN) accelerator, and a stream cipher primitive.

5. PEER-REVIEWED PUBLICATIONS & INTELLECTUAL PROPERTY

- **Refereed International Conference Paper (Lead Author & Presenter):** Arunjyoti Sarma, Kuntal Kumar Das, Dr. L. Oppili Prasad, Dr. Ashutosh Kumar Dubey, *"Piezoelectric Nanogenerator-Based Energy Harvesting System for Low-Power Bioelectronics,"* **Proceedings of the 2026 IEEE 8th Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (IEEE-ECBIOS 2026)**, Tainan, Taiwan, October 2026. (Joint Research: Dept. of Electronics Engg. & Dept. of Ceramic Engg., IIT (BHU) Varanasi).
- **Patented Architecture & Invention Disclosure (Sole Inventor):** Arunjyoti Sarma, *"A Reconfigurable, Power-Gated Universal Shift Register with Multi-Stride Barrel Shifting and Built-In Self-Test for Bit-Serial Neural Acceleration and Cryptographic Primitives (USR-X),"* Indian & International PCT Patent Specification, Ref: USR-X/IIT-BHU/2025. (RTL verification & silicon simulation closed).
- **National Conference Technical Monograph (Lead Designer):** Arunjyoti Sarma, Swati Gupta, Updesh Kumar (Team AstraVortex, IIT (BHU) Varanasi), *"Processing-in-Memory (PIM) Hardware Accelerator Integrated with Indigenous VEGA RISC-V Processor for Real-Time Edge Cancer Diagnostics,"* **DVCon India 2025 Design Contest Technical Monograph**, Bangalore, India.
- **Journal Manuscript in Preparation (Targeted Submission):** Arunjyoti Sarma, Dr. L. Oppili Prasad, *"Multi-Physical Strain Dynamics and Electromechanical Optimization of Serpentine Interconnects on Elastomeric Substrates for Continuous Structural Health Monitoring,"* Manuscript in preparation for submission to **IEEE Sensors Journal**.

6. TECHNICAL SKILLS & CORE COMPETENCIES

- **FPGA & Digital Hardware Design:** Verilog HDL, SystemVerilog, AMD Vivado Design Suite, Spartan-7 / Artix-7 / Kintex-7, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Fixed-Point DSP-Free Arithmetic, RTL Synthesis, Yosys, Verilator, GTKWave, ModelSim, Hardware-in-the-Loop (HIL) Emulation.
- **Electronic Design Automation (EDA) & Simulation:** Cadence Innovus (P&R;, CTS), Cadence Genus (Synthesis), RTL-to-GDSII flow, COMSOL Multiphysics (Solid Mechanics, Electrostatics, Piezoelectric Devices, AC/DC), Tina-TI, LTspice, Proteus, Altium / KiCad PCB Layout.
- **Analog, Sensor & Biomedical Hardware:** Piezoelectric Nanogenerators (PVDF/PENG), Flexible Strain Sensing, Analog Front-End (AFE) Design, Instrumentation Amplifiers, Precision Bridge Readouts, MT3608 DC-DC Converters, Micro-Fabrication & Hot-Press Polymer Processing.
- **Programming & Embedded Firmware:** C, C++ (C++17/20, pybind11), Python (NumPy, SciPy, Pandas, Scikit-learn, PyTorch), MATLAB / Simulink, Embedded C, Microchip PIC, ATmega (Arduino), ARM Cortex-M, ESP32, Raspberry Pi, Linux Shell / PowerShell.
- **Architectures, Protocols & Web Telemetry:** RISC-V ISA (VEGA indigenous core), Processing-in-Memory (PIM), Lattice Cryptography HW, UART, SPI, I2C, AXI4-Lite/AXI4, FTDI High-Speed Serial, TCP/IP, WebSockets, Flask, Docker, Git, LaTeX, Signal Processing, Information Theory, Neural Networks, Real-Time Systems.